MATHEMATICS

QUADRATIC EQUATION
THEORY SHEET
ENGLISH

Piprali Road, Sikar
Ph. 01572-241911, www.matrixedu.in

1. STANDARD FORM OF QUADRATIC EQUATION

An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$ and a, b, c are real numbers

The numbers a, b and c are called the coefficients of the quadratic equation.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The quantity $\mathbf{D = b^2 - 4ac}$ is known as the discriminant of the equation.

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Nature of roots :

1. $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
2. $D = 0 \Leftrightarrow$ roots are real & coincident (equal)
3. $D < 0 \Leftrightarrow$ roots are imaginary.
4. If roots are real then $D \geq 0$

2. RELATION BETWEEN ROOTS AND COEFFICIENTS

If α and β are the roots of quadratic equation $ax^2 + bx + c = 0$, then,

(i) Sum of Roots $\quad S = \alpha + \beta = -\frac{b}{a} = -\frac{\text{Coefficient of x}}{\text{Coefficient of } x^2}$

(ii) Product of Roots $\quad P = \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$

(iii) $|\alpha - \beta| = \left|\frac{\sqrt{D}}{a}\right|$

Notes :

1. Quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in R$ & $a \neq 0$

   Imaginary roots will always occur in conjugate pairs. If one roots is $p + iq$ then the other will be $p - iq$.

2. Quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in Q$ & $a \neq 0$ then ;

   (i) If D is a perfect square, then roots are rational.

   (ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is a surd) then other root will be $p - \sqrt{q}$.

3. SYMMETRIC FUNCTION OF ROOTS

The symmetric expressions of the roots α, β of an equation are those expressions in α and β, which do not change by interchanging α and β. To find the value of such an expression, we generally express that

in terms of $\alpha+\beta$ and $\alpha\beta$.

Some examples of symmetric expressions are–

(i) $\alpha^2+\beta^2$ (ii) $\alpha^2+\alpha\beta+\beta^2$ (iii) $\frac{1}{\alpha}+\frac{1}{\beta}$ (iv) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$

(v) $\alpha^2\beta+\beta^2\alpha$ (vi) $\left(\frac{\alpha}{\beta}\right)^2+\left(\frac{\beta}{\alpha}\right)^2$ (vii) $\alpha^3+\beta^3$ (viii) $\alpha^4+\beta^4$

Notes :

(i) $\alpha^2+\beta^2 = (\alpha+\beta)^2-2\alpha\beta$ (ii) $\alpha^3+\beta^3 = (\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

Q.1 If roots of equation $x^2-5x+3=0$ are α, β then find value of

(i) $\alpha+\beta, \alpha\beta, |\alpha-\beta|$ (ii) $\alpha^2+\beta^2$ (iii) $\alpha^3+\beta^3$

(iv) $\alpha^4+\beta^4$ (v) $\frac{1}{\alpha}+\frac{1}{\beta}$ (vi) $\alpha^2\beta+\beta^2\alpha$

(vii) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ (viii) $(1-\alpha)(1-\beta)$ (ix) $(1+\alpha)(1+\beta)$

Ans. (i) $\alpha+\beta=5 \;;\; \alpha\beta=3 \;;\; |\alpha-\beta|=\sqrt{13}$

(ii) 19
(iii) 80
(iv) 343
(v) 5/3
(vi) 15
(vii) 19/3
(viii) –1
(ix) 9

Sol. (i) Sum fo roots $=\alpha+\beta=\frac{-b}{a}=\frac{-(-5)}{1}\Rightarrow\alpha+\beta=5$

product of roots $=\alpha\beta=\frac{c}{a}=\frac{3}{1}\Rightarrow\alpha\beta=3$

$$|\alpha-\beta|=\left|\frac{\sqrt{D}}{a}\right|=\left|\frac{\sqrt{25-9\times3}}{1}\right|\Rightarrow|\alpha-\beta|=\sqrt{13}$$

(ii) $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=(5)^2-2(3)$

$=19$

(iii) $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$=(5)^3-3\times3\times5$

$=80$

(iv) $\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2(\alpha\beta)^2$

$=(19)^2-2(3)^2$

$=343$

(v) $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{5}{3}$

(vi) $\alpha^2\beta + \beta^2\alpha = \alpha\beta(\alpha + \beta)$

$= 3 \times 5 = 15$

(vii) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\Rightarrow\frac{\alpha^2+\beta^2}{\alpha\beta}$

$\Rightarrow\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

$\Rightarrow\left(\frac{5^2-2\times3}{3}\right)=\frac{19}{3}$

(viii) $(1 - \alpha)(1 - \beta) = 1 - (\alpha + \beta) + \alpha\beta$

$= 1 - 5 + 3$

$= -1$

(ix) $(1 + \alpha)(1 + \beta) = 1 + \alpha + \beta + \alpha\beta$

$= 1 + 5 + 3$

$= 9$

Q.2 If roots of equation $2x^2 - 3x + 5 = 0$ are α, β then find value of.

(i) $\alpha + \beta, \alpha\beta, |\alpha - \beta|$ (ii) $\alpha^2 + \beta^2$ (iii) $\alpha^3 + \beta^3$ (iv) $\alpha^4 + \beta^4$ (v) $\frac{1}{\alpha}+\frac{1}{\beta}$

(vi) $\alpha^2\beta + \beta^2\alpha$ (vii) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$ (viii) $(1-\alpha)(1-\beta)$ (ix) $(1+\alpha)(1+\beta)$

Ans. (i) $\alpha+\beta=\frac{3}{2}$; $\alpha\beta=\frac{5}{2}$; $|\alpha-\beta|=\frac{\sqrt{31}i}{2}$

(ii) $-\frac{11}{4}$ (iii) $-\frac{63}{8}$ (iv) $-\frac{79}{16}$ (v) $\frac{3}{5}$

(vi) $\frac{15}{4}$ (vii) $-\frac{11}{10}$ (viii) 2 (ix) 5

Sol. (i) $\alpha+\beta=\frac{3}{2},\alpha\beta=\frac{5}{2},|\alpha-\beta|=\left|\frac{\sqrt{D}}{a}\right|$

$=\frac{\sqrt{31}i}{2}$

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$=\left(\frac{3}{2}\right)^2-2\left(\frac{5}{2}\right)=-\frac{11}{4}$$

(iii) $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

$$=\left(\frac{3}{2}\right)^3-3\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)$$

$$=\frac{-63}{8}$$

(iv) $\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2(\alpha\beta)^2$

$$=\left(\frac{-11}{4}\right)^2-2\left(\frac{5}{2}\right)^2$$

$$=-\frac{79}{16}$$

(v) $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{5}$

(vi) $\alpha^2\beta+\beta^2\alpha=\alpha\beta(\alpha+\beta)$

$$=\frac{5}{2}\times\frac{3}{2}=\frac{15}{4}$$

(vii) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}$

$$=\frac{\left(-\frac{11}{4}\right)}{\left(\frac{5}{2}\right)}=\frac{-11}{10}$$

(viii) $(1-\alpha)(1-\beta)=1-(\alpha+\beta)+\alpha\beta$

$$=1-\frac{3}{2}+\frac{5}{2}=2$$

(ix) $(1+\alpha)(1+\beta)=1+\alpha+\beta+\alpha\beta$

$$=1+\frac{3}{2}+\frac{5}{2}=5$$

Q.3 If one root of equation $x^2-7x+k=0$ is double of other then find k.

Ans $\frac{98}{9}$

Sol. $x^2-7x+k=0$ $\begin{matrix}\nearrow\ \alpha\\ \searrow\ 2\alpha\end{matrix}$

Sum $\equiv \alpha + 2\alpha = 7$, Product $= (\alpha)(2\alpha) = k$

$\alpha = \frac{7}{3}$ $\qquad$ $k = 2\left(\frac{7}{3}\right)^2$

$k = \frac{98}{9}$

Q.4 If one root of equation $x^2 - 11x + k = 0$ is 3 more than other root then find k.

Ans 28

Sol. $x^2 - 11x + k = 0 \begin{cases} \alpha \\ \alpha + 3 \end{cases}$

Sum $\equiv \alpha + \alpha + 3 = 11$; Product $= (\alpha)(\alpha + 3) = k$

$\alpha = 4$ $\qquad$ $K = 4(4 + 3)$

$K = 28$

Q.5 If one root of equation $x^2 - 6x + k = 0$ is square of other root then find k.

Ans 8

Sol. $\alpha^2 + \alpha = 6$

$\alpha^2 + \alpha - 6 = 0$

$\alpha = -3, 2$

$k = (-3)(9) = -27$ $\qquad$ or $k = (2)(4) = 8$

Q.6 If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

यदि समीकरण $3x^2 + \lambda x - 1 = 0$ के मूल α तथा β के व्युत्क्रमों के वर्गों का योगफल 15 है, तो $6(\alpha^3 + \beta^3)^2$ बराबर है :

Ans. 24

Sol. $3x^2 + \lambda x - 1 = 0$

$\alpha + \beta = \frac{-\lambda}{3}$, $\qquad$ $\alpha\beta = \frac{-1}{3}$

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 15$

$\frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = 15$

$\frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2} = 15$

$\Rightarrow \lambda^2 = 9$ $\qquad$ $\Rightarrow \lambda = \pm 3$

$6\left(\alpha^3+\beta^3\right)^2 = 6\left(\left(\alpha+\beta\right)^3 - 3\alpha\beta\left(\alpha+\beta\right)\right)^2$

$= 6(4) = 24$

4. FORMATION OF QUADRATIC EQUATION

A quadratic equation whose roots are α and β is given by

$(x - \alpha)(x - \beta) = 0$

$\therefore \quad x^2 - \alpha x - \beta x + \alpha\beta = 0$

$\therefore \quad x^2 - (\alpha + \beta)x + \alpha\beta = 0$

i.e $x^2 -$ (Sum of Roots)x + Product of Roots = 0

Q.1 Form a quadratic equation whose roots are 2 & 5.

Ans. $x^2 - 7x + 10 = 0$

Sol. $x^2 - (2 + 5)x + 2 \times 5 = 0$

$x^2 - 7x + 10 = 0$

Q.2 Form a quadratic equation whose coefficients are real & one root is $2 + i$.

Ans. $x^2 - 4x + 5 = 0$

Sol. $\alpha = 2 + i, \quad \beta = 2 - i$

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2 - 4x + 5 = 0$

Q.3 Form a quadratic equation whose coefficients are rational & one root is $2 + \sqrt{3}$.

Ans. $x^2 - 4x + 1 = 0$

Sol. $\alpha = \alpha + \sqrt{3}, \quad \beta = 2 - \sqrt{3}$

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2 - 4x + 1 = 0$

Q.4 If roots of quadratic equation $x^2 - 5x + 3 = 0$ are α & β then form a quadratic equation whose roots are

(i) 2α & 2β (ii) $\alpha + 1, \beta + 1$ (iii) $\frac{1}{\alpha}, \frac{1}{\beta}$ (iv) α^2, β^2 (v) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

Ans. (i) $x^2 - 10x + 12 = 0$ (ii) $x^2 - 7x + 9 = 0$ (iii) $3x^2 - 5x + 1 = 0$ (iv) $x^2 - 19x + 9 = 0$

(v) $3x^2 - 19x + 3 = 0$

Sol. (i) sum of roots $= 2\alpha + 2\alpha = 2(\alpha + \beta) = 10$

Product of roots $= (2\alpha)(2\beta) = 4\alpha\beta = 12$

eq. $x^2 -$ (sum of roots)x + product $= 0$

$x^2 - 10x + 12 = 0$

(ii) $x^2 - (\alpha + 1 + \beta + 1)x + (\alpha + 1)(\beta + 1) = 0$

$x^2 - (\alpha + \beta + 2)x + (\alpha\beta + \alpha + \beta + 1)$

$x^2 - (7)x + 9 = 0$

(iii) $x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha} \times \frac{1}{\beta} = 0$

$x^2 - \left(\frac{\alpha + \beta}{\alpha\beta}\right)x + \frac{1}{\alpha\beta} = 0$

$x^2 - \frac{5}{3}x + \frac{1}{3} = 0$

(iv) $x^2 - (\alpha^2 + \beta^2)x + (\alpha\beta)^2 = 0$

$x^2 - [(\alpha + \beta)^2 - 2\alpha\beta]x + (\alpha\beta)^2 = 0$

$x^2 - [25 - 2 \times 3]x + 9 = 0$

$x^2 - 19x + 9 = 0$

(v) $x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 0$

$x^2 - \frac{\alpha^2 + \beta^2}{\alpha\beta}x + 1 = 0$

$x^2 - \frac{19}{3}x + 1 = 0$

Q.5 If $x^2 + \frac{1}{x^2} = 14; (x > 0)$ find

(i) $x + \frac{1}{x}$ (ii) $x^3 + \frac{1}{x^3}$ (iii) $x^5 + \frac{1}{x^5}$

Ans. (i) 4 (ii) 52 (iii) 724

Sol. (i) $\left(x + \frac{1}{x}\right)^2 - 2 = 14$

$\left(x + \frac{1}{x}\right)^2 = 16$

$\Rightarrow x + \frac{1}{x} = 4 (x > 0)$

(ii) For $x^3 + \frac{1}{x^3}$, $x + \frac{1}{x} = 4$

Cubing both side, we get

$$x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = 64$$

$$x^3 + \frac{1}{x^3} + 3(4) = 69$$

$$\rightarrow x^3 + \frac{1}{x^3} = 52$$

(iii) $$\left(x^2 + \frac{1}{x^2}\right)\left(x^3 + \frac{1}{x^3}\right) = 14 \times 52$$

$$x^5 + \frac{1}{x} + x + \frac{1}{x^5} = 728$$

$$x^5 + \frac{1}{x^5} = 728 - \left(x + \frac{1}{x}\right)$$

$$x^5 + \frac{1}{x^5} = 724$$

Note :

1. If roots of equation $ax^2 + bx + c = 0$ are α & β then $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

Q.6 If roots of equation $x^2 - 4x + 2 = 0$ are α & β then find

(i) $(1 - \alpha)(1 - \beta)$

$x^2 - 4x + 2 = (x - \alpha)(x - \beta)$ .....(1)

Put $x = 1$ in (1)

$1 - 4 + 2 = (1 - \alpha)(1 - \beta) = -1$

(ii) $(1 + \alpha)(1 + \beta)$

Put $x = -1$ in (1)

$1 + 4 + 2 = (-1 - \alpha)(-1 - \beta)$

$(1 + \alpha)(1 + \beta) = 7$

2. Roots of equation $ax^3 + bx^2 + cx + d = 0$ are α, β & γ then $ax^3 + bx^2 + cx + d = a(x - \alpha)(x - \beta)(x - \gamma)$

Q.7 If roots of quadratic equation are $2x^3 - 5x + 9 = 0$ α, β, & γ then find $(1 - \alpha)(1 - \beta)(1 - \gamma)$.

Ans. 3

Sol. $2x^3 - 5x + 9 = 2(x - \alpha)(x - \beta)(x - \gamma)$

Put $x = 1$

$2(1)^3 - 5(1) + 9 = 2(1 - \alpha)(1 - \beta)(1 - \gamma)$

$3 = (1 - \alpha)(1 - \beta)(1 - \gamma)$

Q.8 If roots of $x^{100} - 2x + 9 = 0$ are $x_1, x_2, \ldots\ldots x_{100}$ then find $(1 - x_1)(1 - x_2) \ldots\ldots (1 - x_{100})$.

Ans. 8

Sol. $(x - x_1)(x - x_2) \ldots\ldots (x - x_{100})$

put $x = 1$,

$1 - 2 + 9 = (1 - x_1)(1 - x_2)\ldots\ldots\ldots(1 - x_{100})$

3. If $y = ax^2 + bx + c$ is a perfect square then $a > 0$ & $D = 0$

4. If sum of coefficient of an equation is zero then are root of that equation will be 1

Q.9 If roots of equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are equal then prove that $2b = a + c$.

Sol. Sum of coefficient $= b - c + c - a + a - b = 0$

$\Rightarrow$One root of equation is (1) and roots are equal

$$\text{sum of roots} = 1 + 1 = \frac{-(1 - a)}{b - c}$$

$$2 = \frac{a - c}{b - c}$$

$2b - 2c = a - c$

$2b = a + c$

Q.10 If roots of equation $x^2 - 4x + 5 = 0$ are α & β then find value of

(i) $\alpha^2 - 4\alpha + 7$ (ii) $(\alpha^2 - 4\alpha + 8)(\beta^2 - 4\beta + 8)$ (iii) $(\alpha^2 - 3\alpha + 6)(\beta^2 - 3\beta + 6)$

Ans. (i) 2 (ii) 9 (iii) 10

Sol. (i) $\alpha^2 - 4\alpha + 7$

$\Rightarrow -5 + 7$

$\Rightarrow 2$

(ii) $(\alpha^2 - 4\alpha + 8)(\beta^2 - 4\beta + 8)$

$(-5 + 8)(-5 + 8)$

$\rightarrow 3 \times 3 = 9$

(iii) $(\alpha^2 - 3\alpha + 6)(\beta^2 - 3\beta + 6)$

We have $\alpha^2 = 4\alpha - 5$ & $\beta^2 = 4\beta - 5$

$\Rightarrow (4\alpha - 5 - 3\alpha + 6)(4\beta - 5 - 3\beta + 6)$

$\Rightarrow (\alpha + 1)(\beta + 1)$

$\Rightarrow \alpha\beta + \alpha + \beta + 1$

$\Rightarrow 5 + 4 + 1 \quad \Rightarrow 10$

Q.11 If roots of $x^2 - 2x + 3 = 0$ are α & β then find value of $\alpha^3 + \alpha^2 - 3\alpha + 14$.

Ans. 5

Sol. $\alpha^2 - 2\alpha + 3 = 0$

$\alpha^2 = 2\alpha - 3$

$\alpha^3 = 2\alpha^2 - 3\alpha$

$\alpha^3 = 2(2\alpha - 3) - 3\alpha$

$\alpha^3 = 4\alpha - 6 - 3\alpha$

$\alpha^3 = \alpha - 6$

Putting the value

$\alpha - 6 + 2\alpha - 3 - 3\alpha + 14$

$\Rightarrow 5$

Q.12 If $x^2 - 6x + 4 = 0$ find value of $x^4 - 12x^3 + 44x^2 - 48x + 22$.

Ans. 6

Sol. $\rightarrow$ Divide $x^4 - 12x^3 + 44x^2 - 48x + 22$ by $x^2 - 6x + 4$

By Remainder theorem

$x^4 - 12x^3 + 44x^2 - 48x + 22$

$= (x^2 - 6x + 4) \times Q(n) + R$

Here $x^2 - 6x + 4 = 0$

5. If roots of equation $ax^2 + bx + c = 0$ are α & β and $S_n = p\alpha^n + q\beta^n$ then $aS_n + bS_{n-1} + cS_{n-2} = 0$

Q.13 Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is :

[JEE(Main)Feb.2021]

माना $x^2 - 6x - 2 = 0$ के मूल α तथा β हैं। यदि $n \geq 1$ के लिए $a_n = \alpha^n - \beta^n$ है, तो $\frac{a_{10} - 2a_8}{3a_9}$ का माना है :

[JEE(Main)Feb.2021]

Ans. D

Sol. $$\frac{\alpha^{10}-\beta^{10}-2\left(\alpha^8-\beta^8\right)}{3\left(\alpha^9-\beta^9\right)}=\frac{\alpha^8\left(\alpha^2-2\right)-\beta^8\left(\beta^2-2\right)}{3\left(\alpha^9-\beta^9\right)}$$

$$=\frac{\alpha^8(6\alpha)-\beta^8(6\beta)}{3\left(\alpha^9-\beta^9\right)}=\frac{6\left(\alpha^9-\beta^9\right)}{3\left(\alpha^9-\beta^9\right)}=\frac{6}{3}=2$$

$\because\ \alpha^2=6\alpha-2=0$

$\alpha^2-2=6\alpha$

$\beta^2-2=6\beta$

Q.14 If a and b are the roots of the equation $x^2-7x-1=0$, then the value of $\frac{a^{21}+b^{21}+a^{17}+b^{17}}{a^{19}+b^{19}}$ is equal to _____.

यदि समीकरण $x^2-7x-1=0$ के मूल a तथा b हैं, तो $\frac{a^{21}+b^{21}+a^{17}+b^{17}}{a^{19}+b^{19}}$ का मान बराबर है।

Ans. 51

Sol. $x^2-7x-1=0<_b^a$

By newton's theorem

$S_{n+2}-7S_{n+1}-S_n=0$

$S_{21}-7S_{20}-S_{19}=0$

$S_{20}-7S_{19}-S_{18}=0$

$S_{19}-7S_{18}-S_{17}=0$

$$\frac{S_{21}+S_{17}}{S_{19}}=\frac{S_{21}+\left(S_{19}-7S_{18}\right)}{S_{19}}$$

$$=\frac{S_{21}+S_{19}-7\left(S_{20}-7S_{19}\right)}{S_{19}}$$

$$=\frac{50S_{19}+\left(S_{21}-7S_{20}\right)}{S_{19}}$$

$$=51\cdot\frac{S_{19}}{S_{19}}=51$$

5. TRANSFORMATION OF ROOTS

Let $ax^2+bx+c=0$ be a quadratic equation with two roots α and β. If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α & β, then this equation can be found by finding α in terms of y. Now as α satisfies given equation, put this α in terms of y directly in the equation.

$y=f(\alpha)$

By transformation, $\alpha = g(y)$

$a(g(y))^2 + b(g(y)) + c = 0$

This is the required equation in y.

Q.1 If roots of quadratic equation $x^2 - 4x + 2 = 0$ are α & β then form a quadratic equation whose roots are

(i) $2\alpha, 2\beta$ (ii) $\alpha + 1, \beta + 1$ (iii) $\frac{1}{\alpha}, \frac{1}{\beta}$ (iv) α^2, β^2

(v) $\alpha^2\beta, \beta^2\alpha$

Ans. (i) $y^2 - 8y + 8 = 0$ (ii) $y^2 - 6y + 7 = 0$ (iii) $2y^2 - 4y + 1 = 0$ (iv) $y^2 - 12y + 4 = 0$

(v) $y^2 - 8y + 8 = 0$

Sol. (i) Let $y = 2\alpha$ and we have $x = \alpha$

$y = 2x$

$x = \frac{y}{2}$

Putting the value of x in givne equation

$\left(\frac{y}{2}\right)^2 - 4\left(\frac{y}{2}\right) + 2 = 0$

$y^2 - 8y + 8 = 0$

(ii) Let $y = \alpha + 1, x = \alpha$

$y = x + 1$

$x = y - 1$

$\Rightarrow (y - 1)^2 - 4(y - 1) + 2 = 0$

$y^2 - 6y + 7 = 0$

(iii) Let $y = \frac{1}{\alpha}, x = \alpha$

$y = \frac{1}{x} \rightarrow x = \frac{1}{y}$

$\Rightarrow \left(\frac{1}{y}\right)^2 - 4\left(\frac{1}{y}\right) + 2 = 0$

$2y^2 - 4y + 1 = 0$

(iv) Let $y = \alpha^2$ & $x = \alpha$

$x^2 = y$

equation is $x^2 - ax + 2 = 0$

$x^2 + 2 = 4x$

$(x^2 + 2)^2 = 16x^2$

$(y + 2)^2 = 16y$

$y^2 - 12y + 4 = 0$

(v) Let $y = \alpha^2\beta$, $x = \alpha$

$y = (\alpha\beta)\alpha$

$y = 2\alpha$

$x = \frac{y}{2}$

$\left(\frac{y}{2}\right)^2 - 4\left(\frac{y}{2}\right) + 2 = 0$

$y^2 - 8y + 8 = 0$

Q.2 If roots of cubic equation $x^3 - 4x^2 + 2 = 0$ are α, β & γ then form a cubic equation whose roots are

(i) $2\alpha, 2\beta, 2\gamma$ (ii) $\alpha + 1, \beta + 1, \gamma + 1$ (iii) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ (iv) $\alpha^2, \beta^2, \gamma^2$

Ans. (i) $y^3 - 8y^2 + 16 = 0$ (ii) $y^3 - 7y^2 + 11y - 3 = 0$

(iii) $2y^3 - 4y + 1 = 0$ (iv) $y^3 - 16y^2 + 16y - 4 = 0$

(i) Let $y = 2\alpha$, $x = \alpha$

$x = \frac{y}{2}$

putting the value of x in given equation

$\left(\frac{y}{2}\right)^3 - 9\left(\frac{y}{2}\right)^2 + 2 = 0$

$y^3 - 8y^2 + 16 = 0$

(ii) Let $y = \alpha + 1$, $x = \alpha$

$x = y - 1$

$(y - 1)^3 - 4(y - 1)^2 + 2 = 0$

$y^3 - 7y^2 + 11\,y - 3 = 0$

(iii) Let $y = \frac{1}{\alpha}, x = \alpha$

$x = \frac{1}{y}$

$\left(\frac{1}{y}\right)^3 - 4\left(\frac{1}{y}\right)^2 + 2 = 0$

$2y^3 - 4y + 1 = 0$

(iv) Let $y = \alpha^2$, $x = \alpha$

$y = x^2 \rightarrow x = \sqrt{y}$

$x^3 - 4x^2 + 2 = 0$

$y\sqrt{y} - 4y + 2 = 0$

$y\sqrt{y} = 4y - 2$

squaring

$y^3 = 16y^2 - 16y + 4$

$y^3 - 16y^2 + 16y - 4 = 0$

6.1 SPECIAL CASES ON ROOTS

Roots of equation $ax^2 + bx + c = 0$ are α & β

$\alpha + \beta = \frac{-b}{a}$, $\alpha\beta = \frac{c}{a}$

1. Exactly one root is zero

Sum $\neq 0$ & Product $= 0$

$\frac{-b}{a} \neq 0$ $\qquad$ $\frac{c}{a} = 0$

$a \neq 0$, $b \neq 0$, $c = 0$

2. Both roots are zero

Sum $= 0$ & Product $= 0$

$\frac{-b}{a} = 0$ $\qquad$ $\frac{c}{a} = 0$

$a \neq 0$, $b = 0$, $c = 0$

Now we will form a quadratic equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$

$x = \alpha \Rightarrow y = \frac{1}{\alpha} = \frac{1}{x}$

$x = \frac{1}{y}$

$a\left(\frac{1}{y}\right)^2 + b\left(\frac{1}{y}\right) + c = 0$

Roots of equation $cy^2 + by + a = 0$ are $\frac{1}{\alpha}, \frac{1}{\beta}$

3. Exactly one of α or β is ∞

$a = 0, \quad b \neq 0$

Example : $x + 3 = 0$ can be considered as a quadratic equation whose one root is ∞

4. Both α, β are ∞

$a = 0, \quad b = 0, \quad c \neq 0$

$3 = 0$ can be considered as a quadratic equation whose both roots are ∞

	Condition	a	b	c
1.	Exactly one root is 0	$\neq 0$	$\neq 0$	0
2.	Both roots are 0	$\neq 0$	0	0
3.	Exactly one root is ∞	0	$\neq 0$	
4.	Exactly two roots are ∞	0	0	$\neq 0$
5.	Identity	0	0	0

6.2 IDENTITY

An equation which is true for every value of the variable within the domain is called an identity , for example : $5(a - 3) = 5a - 15$, $(a + b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in R$.

Note :

1. If $ax^2 + bx + c = 0$ is an identity $\Leftrightarrow a = b = c = 0$
2. A quadratic equation cannot have three or more roots & if it has , it becomes an identity.
3. If $ax^2 + bx + c = 0$ is an identity then it will have ∞ roots.

Q.1 $(p - 1)(p - 2)(p - 3)x^2 + (p - 2)(p - 3)x + (p - 3)(p - 4) = 0$ find p if

(i) Exactly one roots is zero (ii) Exactly one root is ∞

(iii) Both roots are infinity (iv) Equation is an identity

An.s (i) $p = 4$ (ii) $p = 1$ (iii) $p = 2$ (iv) $p = 3$

Sol. (i) $a \neq 0, b \neq 0, c = 0$

$(p - 1)(p - 2)(p - 3) \neq 0$, $(p - 2)(p - 3) \neq 0$, $(p - 3)(p - 4) = 0$

$p \neq 1, 2, 3$ $p \neq 2, 3$ $p = 3, 4$

$p = 4$

(ii) $a = 0, b \neq 0$

$p = 1, 2, 3,$ $p \neq 2, 3$ $\Rightarrow p = 1$

(iii) $a = 0, b = 0, c \neq 0$

$p = 1, 2, 3$; $p = 2, 3$; $p \neq 3, 4$

$\Rightarrow p = 2$

(iv) Equation is an identity

$a = 0, b = 0, c = 0$

$p = 1, 2, 3$; $p = 2, 3$; $p = 3, 4$

$\Rightarrow p = 3$

Q.2 Prove that equation $\frac{(x-a)(x-b)}{(c-a)(c-b)} + \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} = 1$ is an identity.

Sol. put $x = a$, $0 + \frac{(a-b)(a-c)}{(a-b)(a-c)} + 0 = 1$

$1 = 1$

Hence $x = a$ is the root of equation

Put $x = b$, $0 + 0 + 1 = 1$

$1 = 1$

b is root of equation

put $x = c$, $1 + 0 + 0 = 1$

$1 = 1$

c is the root of eq.

we have more than teo toots of quadratic equation

$\Rightarrow$ equation is can Identity

6.3 WRITING GENERAL EQUATION

Q.1 If $\alpha^2 - 5\alpha + 3 = 0, \beta^2 - 5\beta + 3 = 0; \alpha \neq \beta$ find value of $\frac{1}{\alpha} + \frac{1}{\beta}$

Ans. 5/3

Sol. $\frac{1}{\alpha} + \frac{1}{\beta} \Rightarrow \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{3}$

Q.2 $a\alpha^2 + b\alpha + c = 3\alpha^2 + 5\alpha; a\beta^2 + b\beta + c = 3\beta^2 + 5\beta; a\gamma^2 + b\gamma + c = 3\gamma^2 + 5\gamma$; α, β & γ are different then find a,b & c.

Ans. $a = 3, b = 5, c = 0$

Sol. $(a-3)x^2 + bx + c - 5 = 0 \Leftarrow \alpha, \beta, \gamma$

Were we have more than two roots of quadratic

$\Rightarrow$ equation is an Identity

$a - 3 = 0$, $b = 0$, $c - 5 = 0$

$a = 3$ $b = 0$ $c = 5$

7. RATIONAL INEQUALITIES AND INTERSECTION OF INTERVALS

Intervals

$3 < x < 7 \Leftrightarrow x \in (3, 7)$ $\qquad$ $3 < x \leq 7 \Leftrightarrow x \in (3, 7]$

$3 \leq x < 7 \Leftrightarrow x \in [3, 7)$ $\qquad$ $3 \leq x \leq 7 \Leftrightarrow x \in [3, 7]$

($\cong$ open $\qquad$ [$\cong$ closed

Ex. $x > 3 \Leftrightarrow x \in (3, \infty)$ $\qquad$ $x < 4 \Leftrightarrow x \in (-\infty, 4)$

Inequalities

if $a > b$

$a + k > b + k$

$a - k > b - k$

$ka > kb$ if $k > 0$

$ka < kb$ if $k < 0$

if $a > b$

$a^2 > b^2$ if a,b, > 0

How to solve

1. Make RHS zero
2. Make leading coeff. positive
3. Factorise into linear factor
4. Mark zeroes on the number line.
5. Mark signs acc. to the rule given below:

   (a) Mark positive on the right most portion & then move towards left.

   (b) If power of corresponding factor is odd then change the sign in next part.

   (c) If power of corressponding factor is even then put same sign in the next part.
6. Write final answer.

Q.1 Find x such that $3x^2 - 7x + 6 < 0$

Ans $x \in \phi$

Sol. $D = 49 - 72 < 0$

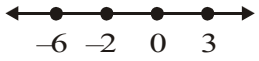
As $D < 0$, $3x^2 - 7x + 6$ will always be positive. Hence $x \in \phi$.

Q.2 $(x^2 - x - 6)(x^2 + 6x) \geq 0$

Ans $x \in (-\infty,-6] \cup [-2,0] \cup [3,\infty)$

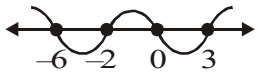
Sol. $(x-3)(x+2)(x)(x+6) \geq 0$

Consider $E = x(x-3)(x+2)(x+6)$, $E = 0 \Rightarrow x = 0, 3, -2, -6$ (all are simple points)



For $x \geq 3$ $E = \underset{+ve}{x}\ \underset{+ve}{(x-3)}\ \underset{+ve}{(x+2)}\ \underset{+ve}{(x+6)}$

= positive



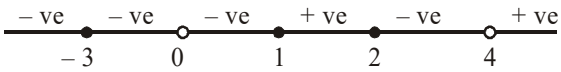
Hence for $x(x-3)(x+2)(x+6) \geq 0$

$x \in (-\infty,-6] \cup [-2,0] \cup [3,\infty)$

Q.3 Find the set of real values of ' x ' such that , $\dfrac{(x-1)^5\,(x-2)^3\,(x+3)^{10}}{x^2\,(x-4)} \leq 0$.

Ans $x \in (-\infty, 0) \cup (0, 1) \cup [2, 4)$

Sol. Domain : $x \in R - \{4, 0\}$



$x \in (-\infty, 0) \cup (0, 1) \cup [2, 4)$

Q.4 Solve the inequality , $\dfrac{3x^2 - 7x + 8}{x^2 + 1} \leq 2$

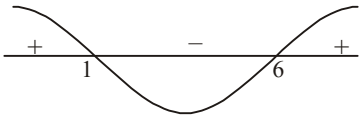
Ans $x \in [1, 6]$

Sol. Domain : $x \in R$

given inequality is equivalent to , $\dfrac{3x^2 - 7x + 8}{x^2 + 1} - 2 \leq 0$

$\Rightarrow \dfrac{3x^2 - 7x + 8 - 2x^2 - 2}{x^2 + 1} \leq 0$

$\Rightarrow \dfrac{x^2 - 7x + 6}{x^2 + 1} \leq 0$



$\Rightarrow \dfrac{(x-1)(x-6)}{x^2 + 1} \leq 0 \qquad \Rightarrow \qquad x \in [1, 6]$

Q.5 Solve the inequality , $\dfrac{1}{x-2} - \dfrac{1}{x} \leq \dfrac{2}{x+2}$.

Ans $x \in \left(-2\,,\, \dfrac{(3-\sqrt{17})}{2}\right] \cup (0\,,\,2) \cup \left[\dfrac{(3+\sqrt{17})}{2}\,,\,+\infty\right)$

Sol. Domain $x \in R - \{-2, 0, 2\}$, given inequality is equivalent to

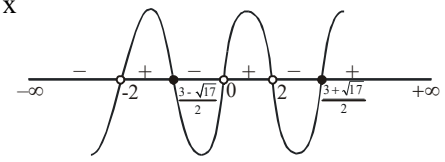
$$\frac{1}{x-2}-\frac{1}{x}-\frac{2}{x+2}\le 0$$

$$\frac{x^2+2x-\left(x^2-4\right)-2\left(x^2-2x\right)}{(x-2)(x+2)x}\le 0$$

$$\frac{\left(x-\frac{\left(3-\sqrt{17}\right)}{2}\right)\left[x-\frac{\left(3+\sqrt{17}\right)}{2}\right]}{(x-2)(x+2)x}\ge 0$$

$$x\in\left(-2,\frac{\left(3-\sqrt{17}\right)}{2}\right]\cup(0,2)\cup\left[\frac{\left(3+\sqrt{17}\right)}{2},+\infty\right)$$

$$\frac{x^2-3x-2}{(x-2)(x+2)x}\ge 0$$



DOUBLE INEQUALITY

Consider inequality $a < b < c$

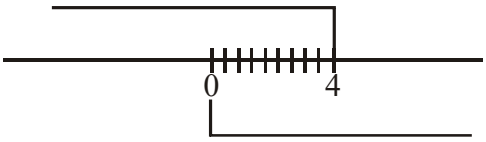
$a < b$ & $b < c$

↓ ↓

A_1 A_2

Ans. $A_1 \cap A_2$

Example – $x \in (-\infty, 4)$ $x \in (0, \infty)$



$x \in (0, 4)$

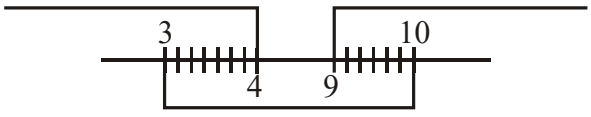
Q.1 $1 < x^2 - 13x + 37 < 7$

$1 < x^2 - 13x + 37$ & $x^2 - 13x + 37 < 7$

$x^2 - 13x + 36 > 0$ $x^2 - 13x + 30 < 0$

$(x-4)(x-9) > 0$ $(x-3)(x-10) < 0$

$x \in (-\infty, 4) \cup (9, \infty)$ $x \in (3, 10)$



$x \in (3, 4) \cup (9, 10)$

8. CONDITION OF COMMON ROOT

8.1 BOTH ROOTS ARE COMMON

For quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ if both roots are common :

Then required conditions is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

8.2 ONE ROOT IS COMMON

Only One Root is Common : Let α be the common root of quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ then

$\therefore\ a_1\alpha^2 + b_1\alpha + c_1 = 0$

$a_2\alpha^2 + b_2\alpha + c_2 = 0$

By Cross multiplication

$$\frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{a_2c_1 - a_1c_2} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\therefore\ \alpha = \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}, \alpha^2 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1},\ \alpha \neq 0.$$

$\therefore$ The condition for only one root common is $(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$

Q.1 $x^2 - 5x - 3 = 0$; $2x^2 + px + q = 0$ find p & q if both roots are common.

Ans. $p = -10$, $q = -6$

Sol. $x^2 - 5x - 3 = 0$

$2x^2 + px + q = 0$

both roots are common

$$\frac{1}{2} = \frac{-5}{p} = \frac{3}{q}$$

$p = -10$, $q = 6$

Q.2 If one root of $x^2 - 3x + 2 = 0$ & $x^2 + kx + 8 = 0$ is common then find k

Ans. $k = -9, -6$

Sol. $x^2 - 3x + 2 = 0$

$x^2 + kx + 8 = 0$

One roots is common cat α is common toots then α satisfy both the equation

$\alpha^2 - 3\alpha + 2 = 0$

$\alpha^2 + k\alpha + 8 = 0$

Applying gross multiplication method

$$\frac{\alpha^2}{-24 - 2k} = \frac{\alpha}{2 - 8} = \frac{1}{k + 3}$$

$$\alpha = \frac{24+2k}{6}, \alpha = \frac{-6}{k+3}$$

$$\frac{12+k}{3} = \frac{-6}{k+3}$$

$k^2 + 15k + 36 = -18$

$k^2 + 15k + 54 = 0$

$k = -6, -9$

Q.3 If one root of $x^2 + (k-29)x - k = 0$, $2x^2 + (2k-43)x + k = 0$ is common then find k

Alternate language

$P_1(x) = x^2 + (k-29)x - k$

$P_2(x) = 2x^2 + (2k-43)x + k$

If P_1 & P_2 have a common linear factor then find k

Another alternate language

If P_1 & P_2 are factors of a cubic polynomial then find k.

(P_1 & P_2 have atleast one common factor)

Ans. $k = 0, 30$

Sol. $x^2 + (k-29)x - k = 0$

$2x^2 + (2k-43)x + k = 0$

by gross multiplication

$$\frac{\alpha^2}{k^2 - 29k + 2k^2 - 43k} = \frac{\alpha}{-2k-k} = \frac{1}{2k-43-2k+50}$$

$$\alpha = \frac{3k^2 - 72k}{-3k} = \frac{-3k}{15}$$

$(k^2 - 24k)(15) = 3k^2$

$5k^2 - 120k - k^2 = 0$

$4k^2 - 120k = 0$

$4k(k-30) = 0$

$k = 0, 30$

Q.4 $x^2 - 5x + 7 = 0; 2x^2 + px + q = 0$, $p, q \in R$, find p & q if equation have a common root.

Ans. $p = -10, q = 14$

Sol. Discriminant of equation $x^2 - 5x + 7 = 0$ is –ve

$\Rightarrow$ Roots are complex coll complex roots are occur in conjugate pair.

Therefore both roots are common

$$\frac{2}{1}=\frac{p}{-5}=\frac{q}{7}$$

$p = -10, q = 14$

Q.5 $a(b-c)x^2 + b(c-a)x + c(a-b) = 0;\ 2x^2 + kx + 9 = 0$ if first equation has equal roots & both equation have a common root then find k.

Ans. $k = -11$

Sol. Sum of coefficients $= a(b-c) + b(c-a) + c(a-b)$

$20 \Rightarrow$ One root of equation is one and given that both roots are equal Therefore other roots is also one.

$2x^2 + kx + 9 = 0$

$2 + k + 9 = 0 \quad \Rightarrow k = -11$

Q.6 $x^2 + bx + c = 0$ & $x^2 + cx + b = 0$ $(b \neq c)$ have a common root then.

(i) Prove that $b + c = -1$

(ii) Prove that equation formed by their uncommon roots will be $x^2 + x + bc = 0$

Sol. $x^2 + bx + c = 0$

$x^2 + cx + b = 0$

$(b-c)x = b - c$

$x = 1, \quad b \neq c$

common roots is (1)

$x^2 + bx + c = 0 \begin{cases} 1 \\ \alpha \end{cases}$; $x^2 + cx + b = 0 \begin{cases} 1 \\ \beta \end{cases}$

product of roots $= \alpha = c$ product of roots $= \beta = b$

eq. whose roots are α & β is

$x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2 - (b + c)x + bc = 0$

Q.7 If one root of equation $a_1x^2 + b_1x + c_1 = 0$ is double of one root of equation $a_2x^2 + b_2x + c_2 = 0$ then find condition.

Ans. $\dfrac{2b_1c_2 - b_2c_1}{c_1a_2 - 4a_1c_2} = \dfrac{c_1a_2 - 4a_1c_2}{4a_1b_2 - 2a_2b_1}$

Sol. If α is the roots of eq. $a_2x^2 + b_2x + c_2 = 0$ then

2α is the root of eq. $a_1x^2 + b_1x + c_1 = 0$

$\Rightarrow 4a_1\alpha^2 + 2b_1\alpha + c_1 = 0$

$c_1\alpha^2 + b_2\alpha + c_2 = 0$

By gross multiplication

$$\frac{\alpha^2}{2b_1c_2 - 4b_2} = \frac{\alpha}{a_2c_1 - 4a_1c_2} = \frac{1}{4a_1b_2 - 2b_1a_2}$$

$$\alpha = \frac{2b_1c_2 - c_1b_2}{a_2c_1 - 4a_1c_2} = \frac{a_2c_1 - 4a_1c_2}{4a_1b_2 - 2b_1a_2}$$

Q.8 If equation $x^2 + ax + 12 = 0, x^2 + bx + 15 = 0, x^2 + (a+b)x + 36 = 0$, have a common negative root find a & b.

Ans. a = 7, b = 8

Sol. Let α is a common roots then

$x^2 + ax + 12 = 0$ .....(1)

$x^2 + bx + 15 = 0$ .....(2)

$x^2 + (a + b)x + 36 = 0$ .....(3)

(1) (2) – (3)

$a^2 - 9 = 0$

$a = -3$

putting value of α in eq. (1) and (2)

we get a = 7, b = 8

9.1 POLYNOMIAL FUNCTION

Standard form

$f(x) = a_0x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots\ldots\ldots\ldots + a_n$

Note:-

1. All powers of x are non-negative integers

eg. $f(x) = 4x^7 + 5x^2 + 3$ yes $f(x) = x^2 + 3x + 4$ yes

$f(x) = x^2 + \frac{1}{x}$ No $f(x) = x + \sqrt{x}$ No

2. Highest power of x is called degree of polynomial (i.e. n).

3. Coefficient of highest power of x is called leading coefficient i.e. a_0

4. $a_0 \neq 0$

5. a_n = constant term/ absolute term/ term independent of x

n = 1 linear $f(x) = ax + b$

n = 2 quadratic $f(x) = ax^2 + bx + c$

n = 3 cubic $f(x) = ax^3 + bx^2 + cx + d$

6. Monic Polynomial : Polynomial whose leading coefficient is 1

9.2 ZEROES OF A POLYNOMIAL

1. Zeroes of a function are those values of x for which value of y is zero.

2. To find zeroes, put y = 0 in the expression.

3. Zeroes are the points on the graph which lie on the x-axis.

Example (1) $f(x) = x - 7$

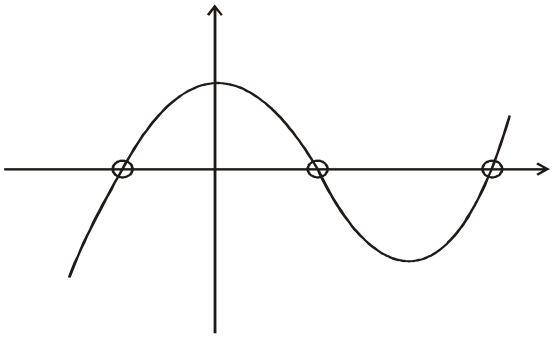
$y = x - 7 = 0$ $x = 7$ (7, 0)

Example (2) $f(x) = x^2 - 6x + 8$

$y = x^2 - 6x + 8 = 0$ $(x - 2)(x - 4) = 0$

$x = 2, 4$ (2, 0) & (4, 0)

Example (3) 3 zeroes of the function



9.3 QUADRATIC POLYNOMIAL AND ITS GRAPHS :

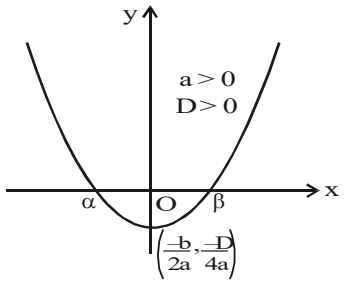
Consider the quadratic expression, $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in R$ then ;

(a) The graph between x, y is always a parabola. If a > 0 then the shape of the parabola is concave upwards & if a < 0 then the shape of the parabola is concave downwards.

(b) The graph of $y = ax^2 + bx + c$ can be divided in 6 broad categories which are as follows :

(Let the real roots of quadratic equation $ax^2 + bx + c = 0$ be α & β where $\alpha \leq \beta$).

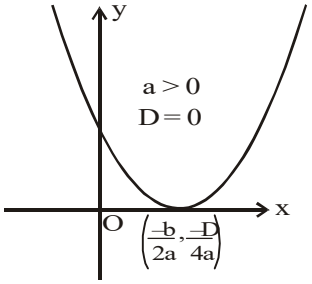
Graph 1 **Graph2** **Graph 3**



Roots are real & distinct

$ax^2 + bx + c > 0 \ \forall \ x \in (-\infty, \alpha) \cup (\beta, \infty)$

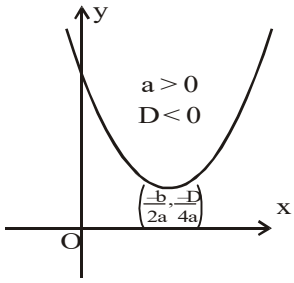
$ax^2 + bx + c < 0 \ \forall \ x \in (\alpha, \beta)$



Roots are coincident

$ax^2 + bx + c > 0 \ \forall \ x \in R - \{\alpha\}$

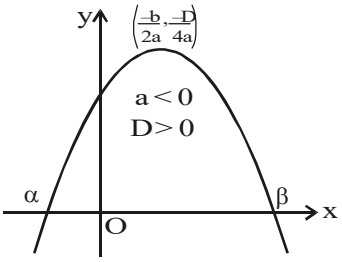
$ax^2 + bx + c = 0$ for $x = \alpha = \beta$



Roots are complex conjugate

$ax^2 + bx + c > 0 \ \forall \ x \in R$

Graph 4

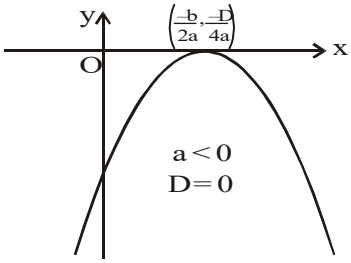


Roots are real & distinct

$ax^2 + bx + c > 0 \ \forall \ x \in (\alpha, \beta)$

$ax^2 + bx + c < 0 \ \forall \ x \in (-\infty, \alpha) \cup (\beta, \infty)$

Graph 5

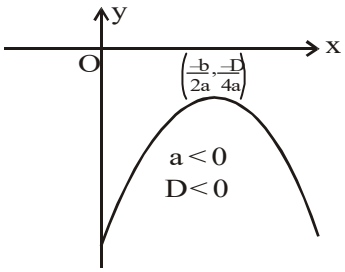


Roots are coincident

$ax^2 + bx + c < 0 \ \forall \ x \in R - \{\alpha\}$

$ax^2 + bx + c = 0$ for $x = \alpha = \beta$

Graph 6



Roots are complex conjugate

$ax^2 + bx + c < 0 \ \forall \ x \in R$

9.4 HOW TO CONSTRUCT GRAPH OF A QUADRATIC POLYNOMIAL

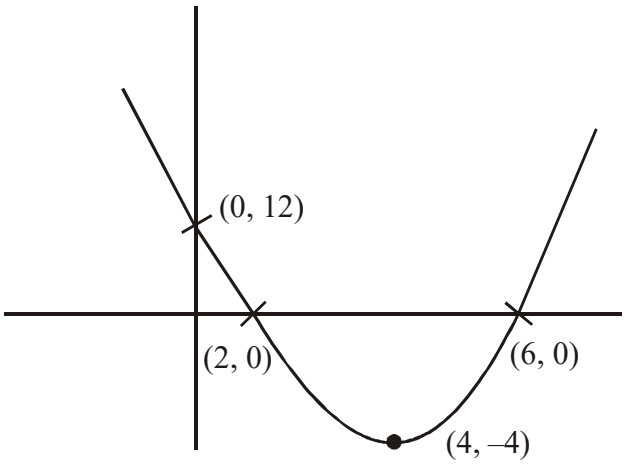
Q.1 Construct graph of $y = x^2 - 8x + 12$

→ Compute a & D $a = 1, \ D = 64 - 48 = 16$

→ Find zeroes (if possible) $0 = x^2 - 8x + 12$

$x = 2, 6$ $(2, 0)$ & $(6, 0)$

→ Find vertex $= \left(\frac{-b}{2a}, -\frac{D}{4a}\right) = (4, -4)$



2. $y = x^2 - 4x - 5$ 3. $y = x^2 + 2x - 3$

4. $y = x^2 + 6x + 9$ 5. $y = 2x - x^2$

6. $y = 9 - x^2$ 7. $y = x^2 + 2x + 5$

9.5 READING GRAPH OF QUADRATIC POLYNOMIAL

The value of expression $f(x) = ax^2 + bx + c$ at $x = x_0$ is equal to y–co–ordinate of the point on parabola $y = ax^2 + bx + c$ whose x–co–ordinate is x_0. Hence if the point lies above the x–axis for some $x = x_0$, then $f(x_0) > 0$ and vice–versa.

Important points

1. $ax^2 + bx + c > 0 \quad \forall\, x \in R \Rightarrow \quad a > 0 \;\&\; D < 0$
2. $ax^2 + bx + c < 0 \quad \forall\, x \in R \Rightarrow \quad a < 0 \;\&\; D < 0$
3. $ax^2 + bx + c \geq 0 \quad \forall\, x \in R \Rightarrow \quad a > 0 \;\&\; D \leq 0$
4. $ax^2 + bx + c \leq 0 \quad \forall\, x \in R \Rightarrow \quad a < 0 \;\&\; D \leq 0$

Note:-

1. $x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$

   Proof: $x^4 + x^2 + 1 = x^4 + 2x^2 + 1 - x^2$

   $= (x^2 + 1)^2 - x^2$

   $= (x^2 + 1 + x)(x^2 + 1 - x)$

2. $\frac{a+b}{2} \geq \sqrt{ab} \quad a, b > 0$

3. Equations of the form $ax^4 + bx^3 + cx^2 \pm bx + a = 0$

Q.1 The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is :

समीकरण $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ के वास्तविक हलों की संख्या है :

Ans. 2

Sol. $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$

Let $f(x) = e^{2x}\left(e^{2x} + \frac{1}{e^{2x}} + 4\left(e^{x} + \frac{1}{e^{x}}\right) - 58\right)$

Let $e^{x} + \frac{1}{e^{x}} = t$

Let $h(t) = t^2 + 4t - 58 = 0$

$t = \frac{-4 \pm \sqrt{16 + 4.58}}{2}$

$\frac{-4 \pm 2\sqrt{62}}{2}$

$t_1 = -2 - 2\sqrt{62}$

$t_2 = -2 + 2\sqrt{62}$ (not possbile)

$t \geq 2$

$e^{x} + \frac{1}{e^{x}} = -2 + 2\sqrt{62}$

$e^{2x} - \left(-2 + 2\sqrt{62}\right)e^{x} + 1 = 0$

$\left(-2 + 2\sqrt{62}\right) - 4$

$4 + 4.62 - 8\sqrt{62} - 4$

$248 - 8\sqrt{62} > 0$

both roots are positive

2 real roots

Q.2 Find number of solutions of the equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$ and how many of them are negative?

समीकरण $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0, x \in R$ के ..... हल हैं जिनमें से ..... ऋणात्मक है ?

Ans. 2,2

Sol. $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^{x} + 1 = 0$

Let $e^{x} = t$

Now $t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

Dividing equation by t^2,

$$t^2+8t+13-\frac{8}{t}+\frac{1}{t^2}=0$$

$$t^2+\frac{1}{t^2}+8\left(t-\frac{1}{t}\right)+13=0$$

$$\left(t-\frac{1}{t}\right)^2+2+8\left(t-\frac{1}{t}\right)+13=0$$

Let $t-\frac{1}{t}=z$

$z^2+8z+15=0$

$(z+3)(z+5)=0$

$z=-3$ ro $z=-5$

So, $t-\frac{1}{t}=-3$ or $t-\frac{1}{t}=-5$

$t^2+3t-1=0$ or $t^2+5t-1=0$

$t=\frac{-3\pm\sqrt{13}}{2}$ or $t=\frac{-5\pm\sqrt{29}}{2}$

as $t=e^x$ so t must be positive,

$t=\frac{\sqrt{13}-3}{2}$ or $\frac{\sqrt{29}-5}{2}$

So, $x=\ln\left(\frac{\sqrt{13}-3}{2}\right)$ or $x=\ln\left(\frac{\sqrt{29}-5}{2}\right)$

Hence two solution and both are negative.

9.6 REMAINDER THEOREM

If we divide a polynomial f(x) by $(x-\alpha)$ the remainder obtained is $f(\alpha)$. If $f(\alpha)$ is 0 then $(x-\alpha)$ is a factor of f(x).

Consider $f(x)=x^3-9x^2+23x-15$

$f(1)=0\Rightarrow$ $(x-1)$ is a factor of f(x).

$f(x)=(x-2)(x^2-7x+9)+3$. Hence $f(2)=3$ is remainder when f(x) is divided by $(x-2)$.

Number = (Quotient) × (factor) + Remainder

Q.1 Find remainder when $x^{100}-7x^2+10$ is divided by $(x-1)$.

Ans 4

Sol. $(x-1)\times Q(n)+R$

put x = 1, $1 - 7 + 10 = R$

$R = 4$

Q.2 Find remainder when $x^{100} - 7x^2 + 10$ is divided by $(x^2 - x)$.

Ans $10 - 6x$

Sol. $x(x - 1) \times Q(n) + ax + b$

put x = 0, $10 = b$

put x = 1, $4 = a + b$

$a = -6$

$R = -6x + 10$

Q.3 A polynomial f(x) when divided by (x –1) gives a remainder of 5 & when divided by (x –4) gives a remainder of 11.

Find remainder when f(x) is divided by (x –1)(x –4).

Ans $2x + 3$

Sol. $f(x) = (x - 1) \times Q_1(x) + 5$

put x = 1, f(1) = 5 .....(1)

$f(x) = (x - 4) \times Q_2(x) + 11$

put x = 4, f(4) = 11 .....(2)

$f(x) = (x - 1)(x - 9) \times Q(x) + ax + b$

put x = 1, f(1) = a + b = 5 .....(3)

put x = 4, f(4) = 4a + b = 11 .....(4)

from (3) and (4), we get a = 2, b = 3

R = 2x + 3

9.7 DESCARTES' RULE OF SIGN

The number of positive real roots is at most the number of sign changes in the sequence of the polynomial's coefficients (omitting zero coefficients), and the difference between the root count and the sign change count is always even. In particular, when the number of sign changes is zero or one, then there are exactly zero or one positive roots.

For negative roots, replace x by $-x$

Example

$f(x) = x^3 + x^2 - x - 1$ has one sign change between the second and third terms, as the sequence of signs is $(+, +, -, -)$. Therefore, it has exactly one positive root.

For negative root $f(-x) = -x^3 + x^2 + x - 1$

This polynomial has two sign changes, as the sequence of signs is $(-, +, +, -)$, meaning that this second polynomial has two or zero positive roots; thus the original polynomial has two or zero negative roots.

JEE Main 2022(June Attempt)

Q.1 The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____.

$x^7 + 5x^3 + 3x + 1 = 0$ के वास्तविक हलों की संख्या है _____.

(1) 0 (2) 1 (3) 3 (4) 5

Sol zero positive and one negative root.

JEE Main 2023(April Attempt)

Q.2 The number of points, where the curve $y = x^5 - 20x^3 + 50x + 2$ crosses the x-axis, is _____.

Ans 5

Sol Check sign of f(1) and f(2) 2 Positive roots

Check sign of f(0), f(-1), f(-2) and f(-3) 3 negative roots

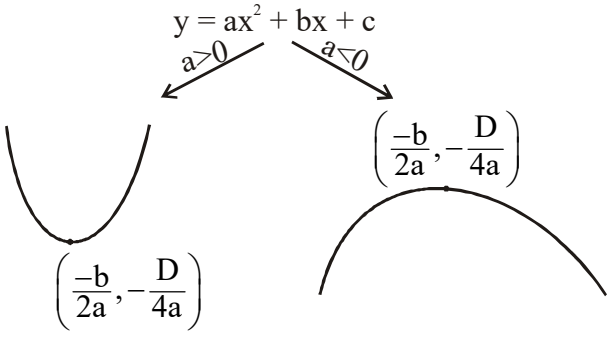
JEE Main 2020(Jan Attempt)

Q.3 The number of real roots of the equation, $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is :

Ans (3) 2

(1) 3 (2) 1 (3) 2 (4) 4

10. RANGE OF QUADRATIC POLYNOMIAL



$y \in \left[-\frac{D}{4a}, \infty\right)$ $y \in \left(-\infty, -\frac{D}{4a}\right]$

Q.1 Find range of $y = x^2 - 2x + 3$

M - I $a = 1 > 0$; $D = 4 - 12 = -8$; $-\frac{D}{4a} = \frac{(-8)}{4} = 2$

$y \in \left[-\frac{D}{4a}, \infty\right) \Rightarrow \quad y \in [2, \infty)$

M-II. $y = x^2 - 2x + 3$

$y = (x - 1)^2 + 2$

$y \in [2, \infty)$

M-III. $y = x^2 - 2x + 3$

$x^2 - 2x + (3 - y) = 0$

$D \geq 0$

$4 - 4(3 - y) \geq 0$

$4 - 12 + 4y \geq 0$

$y \geq 2$

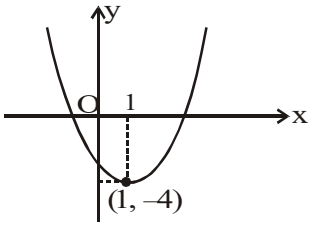
$y \in [2, \infty)$

Q.2 If $y = x^2 - 2x - 3$, then find the range of y when :

(i) $x \in R$ (ii) $x \in [0,3]$ (iii) $x \in [-2,0]$

Ans **(i)** $y \in [-4,\infty)$ (ii) $y \in [-4, 0]$ **(iii)** $y \in [-3, 5]$

Sol. We know that minimum value of y will occur at



$$x = -\frac{b}{2a} = -\frac{(-2)}{2\times 1} = 1$$

$$y_{min} = -\frac{D}{4a} = \frac{-(4+3\times 4)}{4} = -4$$

(i) $x \in R$;

$y \in [-4,\infty)$

Ans.

(ii) $x \in [0, 3]$

$f(0) = -3$, $f(1) = -4$, $f(3) = 0$

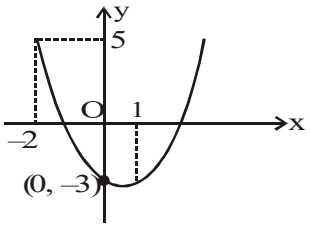
$\because f(3) > f(0)$

$\therefore$ y will take all the values from minimum to f(3).

$y \in [-4, 0]$

Ans.

(iii) $x \in [-2, 0]$



y will take values from f(0) to f(–2)

$f(0) = -3$

$f(-2) = 5$

$y \in [-3, 5]$

11. RANGE OF RATIONAL FUNCTIONS

11.1 $\frac{\text{Linear}}{\text{Linear}}$ (with no common factor)

$$y = \frac{ax + b}{cx + d}$$

$cxy + dy = ax + b$

$cxy - ax = b - dy$

$x = \dfrac{b-dy}{cy-a}$

$y \in R - \left\{\dfrac{a}{c}\right\}$

1. Find range of $y = \dfrac{2x-1}{3x-5}$

Ans $y \in R - \left\{\dfrac{2}{3}\right\}$

2. Find range of $y = \dfrac{4-3x}{x-2}$

Ans $y \in R - \left\{\dfrac{-3}{1}\right\}$

3. Find range of $y = \dfrac{5}{2x+1}$

Ans $y \in R - \{0\}$

4. Find range of $y = \dfrac{x-1}{2x-5}$, $x \geq 0$

Ans $y \in \left(-\infty, \dfrac{1}{5}\right] \cup \left(\dfrac{1}{2}, \infty\right)$

Sol. $2xy - 5y = x - 1$

$2xy - x = 5y - 1$

$x = \dfrac{5y-1}{2y-1} \geq 0$

$y \in \left(-\infty, \dfrac{1}{5}\right] \cup \left(\dfrac{1}{2}, \infty\right)$

5. $y = \dfrac{\sin x - 1}{2\sin x - 5}$

Ans $y \in \left[0, \dfrac{2}{7}\right]$

Sol put $\sin x = t$

$y = \dfrac{t-1}{2t-5}$ $t \in [-1, 1]$

$$t = \frac{5y-1}{2y-1}$$

$$-1 \le \frac{5y-1}{2y-1} \le 1$$

Get $y \in \left[0, \frac{2}{7}\right]$

11.2 If there is common factor

Ques. (i) $y = \frac{2(x-1)}{3(x-1)}$

$$y = \frac{2}{3},\ x \neq 1$$

$$y \in \left\{\frac{2}{3}\right\}$$

(ii) $$y = \frac{(ax+b)(x-\alpha)}{(cx+d)(x-\alpha)}$$

$$y = \frac{ax+b}{cx+b}, x \neq \alpha$$

$$y \in R - \left\{\frac{a}{c}, \frac{a\alpha+b}{c\alpha+d}\right\}$$

Q.1 Find range of $y = \frac{x^2-3x+2}{2x^2+x-3}$

Ans $y \in R - \left\{\frac{1}{2}, \frac{-1}{5}\right\}$

Sol $$y = \frac{(x-1)(x-2)}{(x-1)(2x+3)}$$

$$y = \frac{x-2}{2x+3},\ x \neq 1$$

$$y \in R - \left\{\frac{1}{2}, \frac{-1}{5}\right\}$$

11.3 $\frac{\text{Quadratic}}{\text{Quadratic}}, \frac{\text{Quadratic}}{\text{Linear}}, \frac{\text{Linear}}{\text{Quadratic}}, \frac{1}{\text{Quadratic}}$ (with no common factor)

Steps

1. Cross multiply & form a quadratic in x

2. Now, take two cases

Case-1: Leading coeff $\neq 0$ & $D \geq 0$ solve this inequality & get values of y.

Case-2: Check whether value of y corresponding to leading coeff = 0 will be in range.

3. Now write final answer by taking answers from both cases.

Q.1 $y = \frac{x^2 - x + 1}{x^2 + x + 1}$ Find range of y

Ans $\left[\frac{1}{3}, 3\right]$

Sol **Sol.** $y = \frac{x^2 - x + 1}{x^2 + x + 1} \Rightarrow (y-1)x^2 + (y+1)x + y - 1 = 0$ .....(i)

case-I : if $y \neq 1$, then equation (i) is quadratic in x

and $\because$ x is real

$\therefore D \geq 0$

$\Rightarrow (y+1)^2 - 4(y-1)^2 \geq 0 \Rightarrow (y-3)(3y-1) \leq 0$

$\therefore y \in \left[\frac{1}{3}, 3\right] - \{1\}$

case-II : if $y = 1$, then equation becomes

$2x = 0 \Rightarrow x = 0$ which is possible as x is real.

$\therefore$ Range $\left[\frac{1}{3}, 3\right]$

2. $y = \frac{x^2 + 2x + 3}{x^2 + 2x + 4}$ Find range of y

Ans. $y \in \left[\frac{2}{3}, 1\right)$

Sol. $(x^2 + 2x + 4)y = x^2 + 2x + 3$

$(y-1)x^2 + 2x(y-1) + 4y - 3 = 0$

Case-I

$y - 1 \neq 0$ & $D \geq 0$

$4(y-1)^2 - 4(y-1)(4y-3) \geq 0$

$4(y-1)[y - 1 - 4y + 3] \geq 0$

$(y-1)(-3y+2) \geq 0$ & $(y-1)(3y-2) \leq 0$

$y \in \left[\frac{2}{3}, 1\right]$

Case-II

$y - 1 = 0$ & $y = 1$

$$1 = \frac{x^2 + 2x + 3}{x^2 + 2x + 4}$$

$x^2 + 2x + 4 = x^2 + 2x + 3$

$4 = 3$[Not Possible]

$$y \in \left[\frac{2}{3}, 1\right)$$

3. $$y = \frac{x^2 + 2x - 11}{2x - 6}$$ Find range of y

Ans. $y \in (-\infty, 2] \cup [6, \infty)$

Sol. $y(2x - 6) = x^2 + 2x - 11$

$x^2 - 2x(y - 1) + 6y - 11 = 0$

Here $C \neq 0$ & $D \geq 0$

$4(y - 1) - 4(1)(6y - 11) \geq 0$

$y^2 - 2y + 1 - 6y + 11 \geq 0$

$y^2 - 8y + 12 \geq 0$

$(y - 2)(y - 6) \geq 0$

$$y \in (-\infty, 2] \cup [6, \infty)$$

4. $$y = \frac{x}{x^2 + 1}$$ Find range of y

Ans. $$y \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Sol. $yx^2 - x + y = 0$

Case-I $y \neq 0$ & $D \geq 0$

$1 - 4y^2 \geq 0 \quad \rightarrow 4y^2 - 1 \leq 0$

$$y \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Case-II

$$y = 0 = \frac{x}{x^2 + 1}$$

$x = 0$ (possible)

$$y \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

5. $y=\frac{x^2-x-2}{x^2+3x}$ Find range of y

Ans. $y \in (-\infty, \infty)$

Sol. $x^2y + 3xy = x^2 - x - 2$

$x^2 (y - 1) + n(3y + 1) + 2 = 0$

Case-I $y - 1 \neq 0 \rightarrow y = 1$ & $D \geq 0$

$9y^2 + 6y + 1 - 8(y - 1) \geq 0$

$9y^2 - 2y + 9 \geq 0$

$D < 0 \rightarrow$ +ve $y \in R$

6. $y=\frac{x^2-x-2}{x^2+Kx}$ find K if range of y is $(-\infty, \infty)$

Ans. $k \in (-\infty, -2) \cup (1, \infty)$

Sol. $y=\frac{x^2-x-2}{x^2+kx} \Rightarrow \frac{(x-2)(x+1)}{x(x+k)}, y \in \mathbb{R}$

$K \neq 2, -1$

$x^2y + Kxy = x^2 - x - 2$

$(y - 1)x^2 + x(Ky + 1) + 2 = 0$

Case-I $y - 1 \neq 0$ & $D \geq 0$

$(Ky + 1)^2 - 4(y - 1)(2) \geq 0$

$K^2y^2 + 2ky + 1 - 8y + 8 \geq 0$

$K^2y^2 + 2y(K - 4) + 9 \geq 0 \ \forall \ y \in R$

$D \leq 0$

$4(k - 4)^2 - 4(K^2)(9) \leq 0$

$K^2 - 8K + 16 - 9K^2 \leq 0$

$-8K^2 - 8K + 16 \leq 0$

$K^2 + K - 2 \geq 0$

$K \in (-\infty, -2) \cup (1, \infty)$

12. GENERAL EQUATION OF 2ND DEGREE IN TWO VARIABLES

The general form of a quadratic expression in two variable x & y is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

The condition that this expression may be resolved into two linear rational factors is

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow$ abc + 2 fgh – af^2 – bg^2 – ch^2 = 0 and h^2 – ab > 0

This expression is called discriminant of the above quadratic expression.

Q.1 Factorize $x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$

Ans $(x + y + 1)(x + 2y + 1)$

Sol Method - 1

$x^2 + x(3y + 2) + (2y^2 + 3y + 1) = 0$

$$x = \frac{-(3y+2) \pm \sqrt{(3y+2)^2 - 4(2y^2+3y+1)}}{2}$$

$$x = \frac{-3y + 2 \pm y}{2}$$

$$x = \frac{-2y-2}{2} = -y - 1 \text{ or } x = \frac{-4y-2}{2} = -2y - 1$$

$(x + y + 1)(x + 2y + 1) = 0$

Method - 2.

$x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$

Separate the homogeneous part & factorise it

$x^2 + 3xy + 2y^2$

$x^2 + 2xy + xy + 2y^2$

$(x + y)(x + 2y)$

add c_1 & c_2 to factors

$(x + y + c_1)(x + 2y + c_2)$

$x^2 + 2xy + c_2 x + xy + 2y^2 + c_2 y + c_1 x + 2c_1 y + c_1 c_2$

$x^2 + 3xy + 2y^2 + x(c_1 + c_2) + y(2c_1 + c_2) + c_1 c_2$

Now compare coeffs.

$c_1 + c_2 = 2$

$c_1 c_2 = 1$ & $2c_1 + c_2 = 3$

$c_1 = 1$ & $c_2 = 1$

$(x + y + 1)(x + 2y + 1)$

Q.2 Facotize $x^2 - 3xy + 2y^2 - 2x - 3y - 35 = 0$

Ans. $(x - y + 5)(x - 2y - 7) = 0$

Sol. Step-I Homogenever port

$x^2 - 3xy + 2y^2$

$(x - 2y)(x - y)$

Step-II adding C_1 & C_2

$(x - 2y + C_1)\ (x - y + C_2)$

Step-III Mltiplying & Compare coefficients with given eq.

$x^2 - 3xy + 2y^2 + x(C_1 + C_2) + y(-2C_2 - C_1) + C_1C_2$

Coefficients fo $x = C_1 + C_2 = -2$ .....(1)

$-C_1 - 2C_2 = -3$ .....(2)

$-C_2 = -5 \quad \Rightarrow C_2 = 5 \quad C_1 = -7$

Q.3 If x and y are two real numbers connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, show that x will lie between 3 and 6 and y between 1 and 10.

Sol. Rewrite the given equation as a quadratic in y

i.e., $y^2 + 2y(x - 10) + (9x^2 - 92x + 244) = 0$

Since y is real, the discriminant of this quadratic in y should be ≥ 0

$4(x - 10)^2 - 4\ (9x^2 - 92x + 244) \geq 0$

i.e., $8x^2 - 72x + 144 \leq 0$

i.e., $8(x^2 - 9x + 18) \leq 0$

i.e., $(x - 3)\ (x - 6) \leq 0 \Rightarrow 3 \leq x \leq 6$

Now, rewrite the given equation as a quadratic equation in x.

i.e., $9x^2 + 2x(y - 46) + (y^2 - 20y + 244) = 0$

Since x is real, the discriminant ≥ 0

$4(y - 46)^2 - 36(y^2 - 20y + 244) \geq 0$

$8y^2 - 88y + 80 \leq 0$

i.e., $y^2 - 11y + 10 \leq 0$

i.e., $(y - 1)\ (y - 10) \leq 0 \Rightarrow 1 \leq y \leq 10$

Q.4 $x^2 + 4y^2 - 3x + 2 = 0$ find range of x & y.

Ans. $x \in [1, 2]; y \in \left[-\frac{1}{4}, \frac{1}{4}\right]$

Sol. $x^2 + 4y^2 - 3x + 2 = 0$

For Range of y Quadratic in 'x'

$x^2 + 3x + y^2 + 2 = 0$

$D \geq 0$

$9 - 4(1)(y^2 + 2) \geq 0$

$9 - 4y^2 - 8 \geq 0$

$4y^2 - 1 \leq 0 \rightarrow y \in \left[-\frac{1}{4}, \frac{1}{4}\right]$

Range of x Quadratic in y

$4y^2 + x^2 - 3x + 2 = 0$

$D \geq 0$

$-4(4)(x^2 - 3x + 2) \geq 0$

$x^2 - 3x + 2 \leq 0 \rightarrow (x - 1)(x - 2) \leq 0$

$x \in [1, 2]$

Q.5 $x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$ find range of x & y.

Ans. $x \in (-\infty, \infty); y \in (-\infty, \infty)$

Sol. $a = 1, h = \frac{3}{2}, b = 2, y = 1, f = \frac{3}{2}, \quad c = 1$

$$\Delta = 1 \times 2 \times 1 - 2 \times \frac{3}{2} \times 1 \times \frac{3}{2} - 1 \times \frac{5}{4} - 2 \times 1 - 1 \times \frac{9}{4}$$

$\Delta = 0 \rightarrow$ Range of $x \in R$ & Range of $y \in R$

Q.6 Show that the equation $x^2 + 3xy + 2y^2 + 2x + 3y + 1 = 0$ for every real value of x, there is a real value of y and for every real value of y there is a real value of x.

Sol. $a = 1, h = \frac{3}{2}, b = 2, y = 1, f = \frac{3}{2}, \quad c = 1$

$$\Delta = 1 \times 2 \times 1 - 2 \times \frac{3}{2} \times 1 \times \frac{3}{2} - 1 \times \frac{5}{4} - 2 \times 1 - 1 \times \frac{9}{4}$$

$\Delta = 0 \rightarrow$ Range of $x \in R$ & Range of $y \in R$

13. THEORY OF EQUATION INVOLVING HIGHER DEGREE

S_1 = Sum taken one at a time S_2 = Sum taken two at a time

S_3 = Sum taken three at a time

Example. α, β $S_1 = \alpha+\beta$ $S_2 = \alpha\beta$

Example α, β, γ $S_1 = \alpha+\beta+\gamma$, $S_2 = \alpha\beta+\beta\gamma+\gamma\alpha$, $S_3 = \alpha\beta\gamma$

We can observe S_1 = sum S_{Last} = product

Exm. $a_0 x^{100} + a_1 x^{99} + \ldots\ldots\ldots\ldots + a_{100} = 0$

$$S_1 = -\frac{\text{coeff. of } x^{99}}{\text{L.C.}} \qquad S_2 = +\frac{\text{coeff. of } x^{98}}{\text{L.C.}}$$

$$S_{40} = +\frac{\text{coeff. of } x^{60}}{\text{L.C.}} \qquad S_{73} = -\frac{\text{coeff. of } x^{27}}{\text{L.C.}}$$

Sum of roots = S_1

$$\text{Product of roots} = S_{100} = +\frac{\text{coeff. of } x^0}{\text{L.C.}}$$

Sum of squares of roots :

$(\alpha+\beta)^2 = \alpha^2+\beta^2 + 2\alpha\beta$

$(\alpha+\beta+\gamma)^2 = (\alpha^2+\beta^2+\gamma^2) + 2(\alpha\beta+\beta\gamma+\gamma\alpha)$

$(S_1)^2 = (\sum\alpha^2) + 2\,S_2$

Q.1 If roots of equation $x^3 - 3x + 7 = 0$ α, β & γ are then find

(i) $\alpha + \beta + \gamma,\ \alpha\beta + \beta\gamma + \alpha\gamma,\ \alpha\beta\gamma$ (ii) $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$ (iii) $\alpha^2 + \beta^2 + \gamma^2$

(iv) $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}$ (v) $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ (vi) $\alpha^3 + \beta^3 + \gamma^3$

Ans. (i) $\alpha+\beta+\gamma=0,\ \alpha\beta+\beta\gamma+\gamma\alpha=-3,\ \alpha\beta\gamma=-7$ (ii) $\frac{3}{7}$ (iii) 6

(iv) 0 (v) 9 (vi) –21

Sol. (i) $\alpha+\beta+\gamma=-\frac{0}{1}, \alpha\beta+\beta\gamma+\gamma\alpha=-3, \alpha\beta\gamma=-7$

$= 0$

(ii) $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} \Rightarrow \frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}$

$\Rightarrow \frac{-3}{-7} \Rightarrow \frac{3}{7}$

(iii) $(\alpha^2 + \beta^2 + \gamma^2) = (\alpha + \beta + \gamma)^2 - 2\,(\alpha\beta + \beta\gamma + \gamma\alpha)$

$= 0^2 - 2(-3)$

$= 6$

(iv) $\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha} \Rightarrow \frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} \Rightarrow 0$

(v) $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$

$\Rightarrow (\alpha\beta+\beta\gamma+\gamma\alpha)^2 - 2\left(\alpha\beta^2\gamma+\beta\gamma^2\alpha+\gamma\alpha^2\beta\right)$

$\Rightarrow (\alpha\beta+\beta\gamma+\gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha+\beta+\gamma)$

$\Rightarrow (-3)^2 - 2(-7)(0) = 9$

(vi) If $\alpha + \beta + \gamma = 0$ then

$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$

$= 3(-7) = -21$

Q.2 If roots of equation $x^5 - 7x^4 + 3x^3 + 4x - 5 = 0$ are x_i, i = 1, 2, 3, 4, 5 find $\sum_{i=1}^{5} x_i^2$.

Ans. 43

Sol. We know that

$(s_1)^2 = \sum x_i^2 + 2(s_2)$

$(7)^2 - 2(3) = \sum x_i^2$

$\sum x_i^2 = 49 - 6 \Rightarrow 43$

Q.3 If sum of two roots of equation $x^3 + 4x^2 - 9x - 36 = 0$ is zero. Solve the equation.

Ans. {–3, 3, –4}

Sol. $\alpha + \beta = 0$ (Given)

$S_1 = \alpha + \beta + \gamma = -4$

$\gamma = -4$

$x^3 + 4x^2 - 9x - 36 = (x + 4)(x^2 - 9)$

$x = -9, \qquad x^2 = y \rightarrow x = \pm 3$

Q.4 If one root of equation $x^3 - 6x^2 + Kx - 6 =$ is average of other two roots then find K.

Ans. 11

Sol. $\alpha = \frac{\beta+\gamma}{2}$ (Given), $\qquad x^3 - 6x^2 + Kx - 6 = 0 \begin{matrix} \nearrow \alpha \\ \rightarrow \beta \\ \searrow \gamma \end{matrix}$

$2\alpha = \beta + \gamma \qquad \alpha + \beta + \gamma = 6$

$\alpha + 2\gamma = 6$

$\gamma = 2$

$(2)^3 - 6(2)^2 + K(2) - 6 = 0$

$8 - 24 + 2K - 6 = 0$

$\Rightarrow K = 11$

Q.5 If roots of equation $4x^3 - 24x^2 + 47x - 30 = 0$ are a,b,c find area of ΔABC.

Ans. $\frac{3}{2}$

Sol. $4x^3 - 24x^2 + 47x - 30 = 0$ < a, b, c

$$a + b + c = \frac{24}{4}$$

$a + b + c = 6$

Area $= \sqrt{s(s-a)(s-b)(s-c)}$, Where $S = \frac{a+b+c}{2}$

$2\sqrt{3(3-a)(3-b)(3-c)}$ $\qquad S = \frac{6}{2}$

$S = 3$

We know that $4x^3 - 29x^2 + 47x - 30 = 4(x-a)(x-b)(x-c)$

Put $x = 3$, $\frac{4(3)^3 - 24(3)^2 + 47(3) - 30}{4} = (3-a)(3-b)(3-c)$

$$(3-a)(3-b)(3-c) = \frac{3}{4}$$

Area $= \sqrt{3 \times \frac{3}{4}} \Rightarrow$ Area $= \frac{3}{2}$

14. LOCATION OF ROOTS :

This article deals with an elegant approach of solving problems on quadratic equations when the roots are located / specified on the number line with variety of constraints.

When conditions on roots are given then we will use graphical approach. We will apply conditions on

(1) D

(2) x-coordinate of vertex $\left(\text{i.e.} -\frac{b}{2a}\right)$

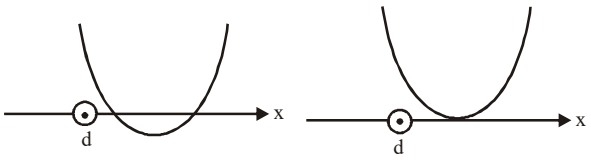
(3) a f (p)

Type-1 :

(a) Both roots of the quadratic equation are greater than a specific number (say d).

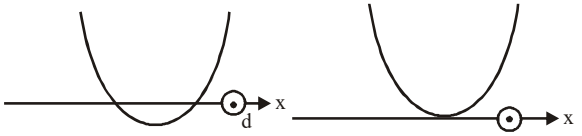
The necessary and sufficient condition for this are :

(i) $D \geq 0$; (ii) $a f(d) > 0$; (iii) $-\frac{b}{2a} > d$



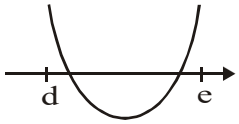
(b) When both roots of the quadratic equation are less than a specific number d than the necessary and sufficient condition will be :

(i) $D \geq 0$; (ii) $a f(d) > 0$; (iii) $-\frac{b}{2a} < d$



Type-2 :

When both roots are confined between the number d and e (d < e).



The necessary and sufficient condition for this are :

(i) $D \geq 0$; (ii) $a f(d) > 0$; (iii) $a f(e) > 0$

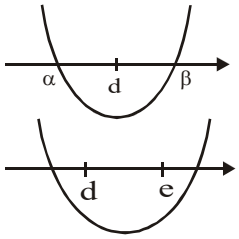
(iv) $d < -\frac{b}{2a} < e$

Note :- If atleast one particular point comes between roots then we will not apply conditions on D & $\frac{-b}{2a}$

Type-3 :

Both roots lie on either side of a fixed number say (d). Alternatively one root is greater than 'd' and other root less than 'd' or 'd' lies between the roots of the given equation.

The necessary and sufficient condition for this are : $a f(d) < 0$



Note : Consideration of discriminant is not needed.

Type-4 :

One root is greater than e and the other roots is less than d (d < e).

The necessary and sufficient condition for this are : $a f(d) < 0$ and $a f(e) < 0$

Q.1 Write conditions if

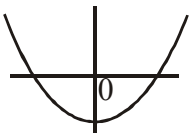
(i) One root is +ve & other is –ve

(ii) Both roots are +ve

(iii) Both roots are –ve

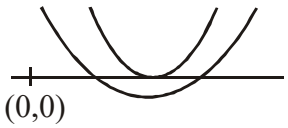
(iv) One root is positive & other is less than –6

Sol (i) One root is +ve & other is –ve



a f (0) < 0

(ii) Both roots are +ve

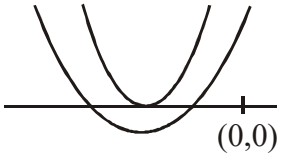


$D \geq 0$

$\frac{-b}{2a} > 0$

a f (0) > 0

(iii) Both roots are –ve

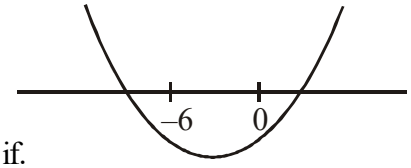


$D \geq 0$

$\frac{-b}{2a} < 0$

a f (0) > 0

(iv) One root is positive & other is less than -6



a f (0) < 0

a f (– 6) < 0

Q.2 Find k if.

(i) Both roots of equation $x^2 - 6kx + (2 - 2k + 9k^2) = 0$ are greater than 3.

(ii) Roots of equation $x^2 - (k + 1)x + k^2 + k - 8 = 0$ lie on either side of 2.

(iii) Both roots of equation $x^2 + 2(k - 3)x + 9 = 0$ lie in (–6, 1)

(iv) One root of equation $(k - 5)x^2 - 2kx + k - 4 = 0$ is less than one & other is greater than 2.

(v) Both roots of equation $x^2 - (k + 2)x + 2k = 0$ are greater than –1.

Ans. (i) $k \in \left(\frac{11}{9}, \infty\right)$ (ii) $k \in (-2, 3)$ (iii) $k \in \left[6, \frac{27}{4}\right)$ (iv) $k \in (5, 24)$

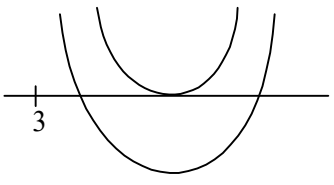
(v) $k \in (-1, \infty)$

Sol. (i) (a) $D \geq 0$

$36k^2 - 4(1)(2 - 2k + 9k^2) \geq 0$

$9k^2 - 2 + 2k - 9k^2 \geq 0$

$k - 1 \geq 0$



(b) $-\frac{(-6k)}{2(6)} > 3$

$k > 1$

(c) af(3) > 0

$9 - 18k + 2 - 2k + 9k^2 > 0$

$9k^2 - 20k + 11 > 0$

$9k^2 - 9k - 11k + 11 > 0$

$9k(k - 1) - 11(k - 1) > 0$

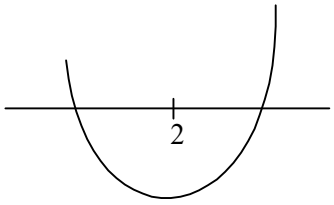
$(k - 1)(9k - 11) > 0$

$K \in (-\infty, 1) \cup \left(\frac{11}{9}, \infty\right)$

(a) $\cap$ (b) $\cap$ (c)

$\Rightarrow k \in \left(\frac{11}{9}, \infty\right)$

(ii) $4f(2) < 0$



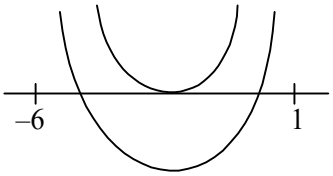
$4 - 2(k + 1) + k^2 + k - 8 < 0$

$k^2 - k - 6 < 0$

$(k - 3)(k + 2) < 0$

$k \in (-2, 3)$

(iii) (a) $D \geq 0$



$4(k - 3)^2 - 4(1)(9) > 0$

$k^2 - 6k + 9 - 9 > 0$

$k \in (-\infty, 0) \cup (6, \infty)$

(b) $-6 < \frac{-2(k-3)}{2} < 1$

$6 > k - 3 > -1$

$9 > k > 2$

(c) $af(-6) > 0$

$36 - 12(k - 3) + 9 > 0$

$12 - 4k + 12 + 3 > 0$

$-4k + 27 > 0$

$k < \frac{27}{4}$

(d) $af(v) > 0$

$1 + 2k - 6 + 9 > 0$

$2k + 9 > 0$

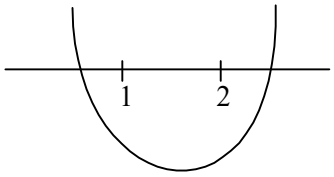
$k > -2$

Intreaction

(a) $\cap$ (b) $\cap$ (c) $\cap$ (d)

$K \in \left[6, \frac{27}{4}\right)$

(iv) (a) af(1) < 0



(k – 5) [k – 5 – 2k + k – 9] < 0

(k – 5) [–9] < 0

k – 5 > 0

k > 5

(b) af(2) < 0

(k – 5) (4k – 20 – 4k + k – 4) < 0

(k – 5) (k – 24) < 0

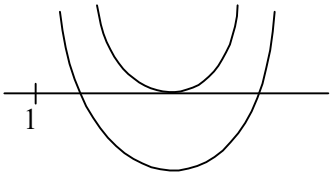
k $\in$ (5, 24)

Intersutin of (a) and (b) k $\in$ (5, 24)

(v) (a) D $\geq$ 0

$(k + 2)^2 - 4(2k) \geq 0$

$k^2 + 4k + 4 - 8k \geq 0$



$k^2 - 4k + 4 \geq 0$

$(k - 2)^2 \geq 0$

k $\in$ R

(b) $\frac{-b}{2a} > -1$

$\frac{k+2}{2} > -1$

k + 2 > – 2

k > –4

(c) a*f*(–1) > 0

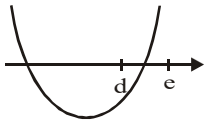
1 + k + 2 + 2k > 0

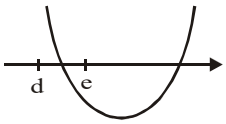
3k + 3 > 0 $\rightarrow$ k $\in$ (–1, ∞)

Note:- If D comes out to be a perfect square then we will directly compute the roots & apply the given condition.

Type-5 :

Exactly one root lies in the interval (d, e).





The necessary and sufficient condition for this are :

$f(d) \cdot f(e) < 0$

Q.3 $x^2 - (k+1)x + 2k = 0$

(i) Find k if exactly one root lies in (0, 3)

$f(0)\, f(3) < 0$

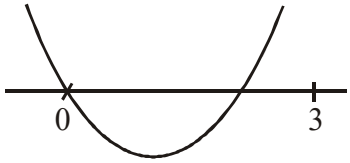
$(2k)(9 - 3k - 3 + 2k) < 0$

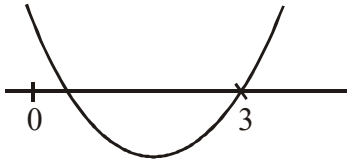
$2k(-k + 6) < 0$

$k(k - 6) > 0$

$k \in (-\infty, 0) \cup (6, \infty)$

Two more graphs are possible.





$k = 0$ $\qquad$ $k = 6$

$x^2 - x = 0$ $\qquad$ $x^2 - 7x + 12 = 0$

$x(x - 1) = 0$ $\qquad$ $(x - 3)(x - 4) = 0$

$x = 0, 1$ $\qquad$ $x = 3, 4$

$k = 0$ $\qquad$ $k = 6$

$k \in (-\infty, 0] \cup (6, \infty)$

(ii) exactly one root lies in [0, 3]

$k = 0$ $\qquad$ $k = 6$

$k \in (-\infty, 0) \cup [6, \infty)$

(iii) exactly one root lies in [0, 3)

$k = 0$ $\qquad$ $k = 6.$

$k \in (-\infty, 0) \cup (6, \infty)$

(iv) exactly one root lies in (0, 3]

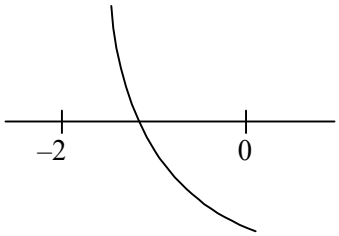
$k = 0$ $\qquad$ $k = 6$

$k \in (-\infty, 0] \cup [6, \infty)$

Q.4 $x^2 + kx + k^2 + 6k = 0$ find k if exactly one root lies in $(-2, 0]$

Ans. $k \in [-6, -2) \cup (-2, 0)$

Sol.

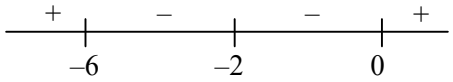


$f(-2)\, f(0) < 0$

$(k^2 + 6k)(4 - 2k + k^2 + 6k) < 0$

$k(k + 6)(k^2 + 4k + 4) < 0$

$k(k + 6)(k + 2)^2 < 0$



$k \in (-6, -2) \cup (-2, 0)$

Chek for

$k = -6$	for $k = -2$	for $k = 0$
$x^2 - 6x = 0$	$x^2 - 2x - 8 = 0$	$x^2 = 0$
$x = 0, 6$	$x = -2, 4$	$x = 0, 0$

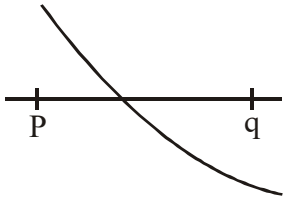
$k \in [-6, -2) \cup (-2, 0)$

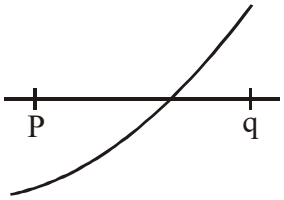
Important Concept

1. $f(x) = ax^2 + bx + c$

If $f(p)$ & $f(q)$ are of opposite signs i.e. $f(p).\, f(q) < 0$

then equation $ax^2 + bx + c = 0$ will have exactly one root in (p, q)





Q.5 If $(a + b + c)(4a + 2b + c) < 0$ then prove that equation $ax^2 + bx + c = 0$ will have real roots.

Sol $f(1)\, f(2) < 0$

then given equation will have exactly one root in (1, 2)

So, equation will have real roots.

Q.6 Prove that equation $(x - a)(x - c) - \lambda(x - b)(x - d) = 0$; $a > b > c > d$

will have real roots $\forall\, \lambda \in R$

Alternate language

If equation $(x-a)(x-c)-\lambda(x-b)(x-d)=0$; $a>b>c>d$

has real roots find λ

Sol $f(b)=(b-a)(b-c)=(-ve)(+ve)=-ve$

$f(d)=(d-a)(d-c)=(-ve)(-ve)=+ve$

$f(b).f(d)<0$.

So, $f(x)$ will have exactly one root in (d, b)

Hence, $f(x)$ will have real roots.

Q.7 $y=\dfrac{(x-a)(x-c)}{(x-b)(x-d)}$ $a>b>c>d$ Prove that range of y is $(-\infty, \infty)$

Sol $y(x-b)(x-d)=(x-a)(x-c)$

$(x-a)(x-c)-y(x-b)(x-d)=0$

$y \in (-\infty, \infty)$

15. TOUGHER PROBLEMS ON LOCATION OF ROOTS

1. $y=ax^2+bx+c$; $a>0$ Find condition(s) if

(i) y is always positive.

(ii) y is always negative.

(iii) $y<0$ for atleast one real x.

2. $ax^2+bx+c=0$; $a>0$ Find condition(s) if

(i) Equation at least one positive real root.

(ii) Equation has atleast negative root.

3. $y=ax^2+bx+c$; $a>0$ Find condition(s) if

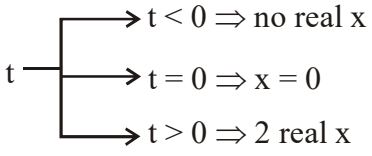
(i) $y>0$ $\forall$ positive x.

(ii) $y<0$ for atleast one negative x.

4. $x^4+x^2(1-2k)+k^2-1=0$

put $x^2=t$

$t^2+t(1-2k)+k^2-1=0$



Case-1 – Quadratic in t has no real roots

implies no real solution in x.

Case-2 – Both roots of quadratic in t are –ve

$\Rightarrow$ no real solution of x

$D \geq 0$

$\frac{-b}{2a} < 0$

$a\, f(0) > 0$

Case-3 – One root is –ve & other one is zero

$\Rightarrow$ one real solution of x

$f(0) = 0$

$\frac{-b}{2a} < 0$

$D > 0$

Case-4 – Both roots are zero

$\Rightarrow$ one real solution of x

$a \neq 0$

$b = 0, c = 0$

Case-5 – One root is –ve & other is +ve

$\Rightarrow$ two real solution of x.

$a\, f(0) < 0$

Case-6 – One root is zero & other is +ve

$\Rightarrow$ three real solution of x.

$f(0) = 0$

$D > 0$

$\frac{-b}{2a} > 0$

Case-7 – Both roots are +ve & equal

$\Rightarrow$ two real solution of x

$\frac{-b}{2a} > 0$

$D > 0$

$a\, f(0) > 0$

Case-8 – Both roots are +ve & distinct

$\Rightarrow$ four real solution of x

$D > 0$

$a\, f(0) > 0$

$\frac{-b}{2a} > 0$

4. Find k if equation $x^4 + x^2 (1 - 2k) + k^2 - 1 = 0$ has.

(i) No real solution of x.

(ii) One real solution of x.

(iii) Two real solutions of x.

(iv) Three real solutions of x.

(v) Four real solutions x.

Ans. (i) $k \in (-\infty, -1) \cup \left(\frac{5}{4}, \infty\right)$ (ii) $k \in \{-1\}$ (iii) $k \in (-1,1) \cup \left\{\frac{5}{4}\right\}$ (iv) $k \in \{1\}$ (v) $k \in (1, 5/4)$

Sol. (i) No real solution of x

(a) Both roots are Imginary

(b) Both roots are –ve

(a) $D < 0$

$(1 - 2k)^2 - 4(k^2 - 1) < 0$

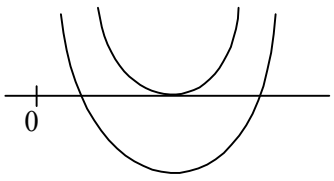
$4k^2 - 4k + 1 - 4k^2 + 4 < 0$

$-4k + 5 < 0$

$k > \frac{5}{4}$

(b) (1) $D \geq 0$

$k \leq \frac{5}{4}$



(2) $\frac{-b}{2a} > 0$

$-(1 - 2k) > 0$

$2k - 1 > 0$

$k > \frac{1}{2}$

(3) $a\, f(0) > 0$

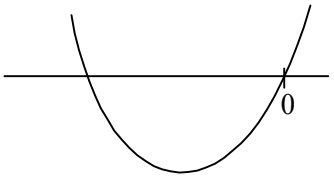
$k^2 - 1 > 0$

$k \in (-\infty, -1) \cup (1, \infty)$

In terssution $k \in (-\infty, -1)$

(a) $\cup$ (b) $\Rightarrow$ $k \in (-\infty, -1) \cup \left(\frac{5}{4}, \infty\right)$

(ii) One real solution of x
(a) one roots is –ve and other is zero
(b) Both roots are zero.
(a) (i) D > 0



$k < \frac{5}{4}$

(ii) a f(0) = 0
$k = \pm 1$

(iii) $\frac{-b}{2a} < 0 \Rightarrow k < \frac{1}{2}$

Intersection k = –1
(b) b = 0 and c = 0

$k = \frac{1}{2}$ $\qquad$ $k^2 - 1 = 0$

$k = \pm 1$

Intersection $k \in \phi$

$k \in \{-1\}$

(iii) Case-I
a f(0) < 0
$k^2 - 1 < 0$
k (–1, 1)
Case-II

D = 0, $\qquad$ $\frac{-b}{2a} > 0$, $\qquad$ a f(0) > 0

$k = \frac{5}{4}$ $\qquad$ $k > \frac{1}{2}$ $\qquad$ $k \in (-\infty, -1)\ (1, \infty)$

Intersection : $k \in (1, \infty)$

Final Ans. $k \in (-1, 1) \cup (1, \infty)$

(iv) D > 0, $\qquad$ a f(0) = 0, $\qquad$ $\frac{-b}{2a} > 0$

$k < \frac{5}{4}$ $\qquad$ $k^2 - 1 = 0$ $\qquad$ $k > \frac{1}{2}$

Intersection : k = –1

(v) D > 0, $\qquad$ $\frac{-b}{2a} > 0$, $\qquad$ a f(0) > 0

$k < \frac{5}{4}$ $\qquad$ $k > \frac{1}{2}$ $\qquad$ $k^2 - 1 > 0$

$k \in (-\infty, -1) \cup (1, \infty)$

Intersection : $k \in \left(1, \frac{5}{4}\right)$